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Deep Learning and Its Applications (CO3133)

Semester 261 – Class CC01 – Group G-02

Assignment 3 – Milestone 3

M3 Final Report

Knowledge-Based Visual Question Answering

Multimodal Deep Learning on A-OKVQA

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1 Problem and data description

1.1 Problem statement

Visual question answering (VQA) asks a model to answer a question about an image. In A-OKVQA the questions also need commonsense or world knowledge: "what does this company specialise in", "which gang wears these colours". Each question comes with 4 choices, so the task is multiple-choice classification over the choices. The prediction unit is one (image, question) pair.

Main difficulties. (1) Two modalities must be combined: the image shows what is there, the question and the choices say what to do with it. (2) Many answers need knowledge that is not visible. (3) The training set is small for a multimodal model (15,346 questions), so we use frozen pretrained encoders. (4) Workers often disagree on the free-form answer (32.4% of questions have low agreement).

Research question. How much does each modality contribute, and does a token-level fusion of image and text help over a simple late fusion of global vectors?

1.2 Source, version and licence

Dataset: A-OKVQA [1], a visual question answering benchmark whose questions need commonsense or world knowledge in addition to what is visible in the image.

https://huggingface.co/datasets/HuggingFaceM4/A-OKVQA

Version: Hugging Face revision `d1b0efa` (`d1b0efa3a436e9101dfbde3752db7607da696c35`), downloaded on 9 November 2026 with `Assignment3/download.py`. Parquet files with the JPEG bytes inside, about 1.2 GB in total.

Licence: the A-OKVQA annotations are released by AllenAI under Apache 2.0. The images are COCO 2017 photos [2], which keep their Flickr licences; we use them only for course work.

How the data was made: crowd workers looked at a COCO image and wrote a question that cannot be answered from the image alone or from knowledge alone. Each question has 4 answer choices, 10 free-form answers from 10 different workers, and 3 short rationales that explain the answer.

1.3 Task, modalities, input and output

Task: multiple-choice VQA. One sample is one image with one question and exactly 4 choices.

Item	Definition
Modality A	Text: the question (mean 8.8 words) and the 4 choice texts (mean 1.3 words)
Modality B	Image: one RGB COCO photo (median 640×480), resized to 224×224
Extra annotation	10 direct answers and 3 rationales per question. Used for evaluation and analysis only, never as model input
Output	Index of the correct choice, $\hat{y} \in \{0, 1, 2, 3\}$
Prediction unit	One (image, question) pair

Table 1: Task definition.

Formulation. For choice j the model gives a score $s_j = f_\theta(\text{image}, \text{question}, \text{choice}_j)$, $p_j = \text{softmax}_j(s_j)$ and $\hat{y} = \arg\max_j p_j$. Training minimises the cross-entropy $-\log p_y$.

Why this task is multimodal. A worker wrote each question while looking at that image, so the pair is linked by the same scene. The answer needs both: the image says *what* is there (a red shirt, a clock at seven), and the question with the choices says what to do with it (which gang wears red, what hour it is). Section 1.12 shows that text alone gets only a few points above chance.

1.4 Size

The handbook asks for about 10,000 question–image pairs for VQA. The official test split (6,702 questions) has no public labels, so we count only labelled pairs: 17,056 (train) + 1,145 (validation) = **18,201 labelled pairs on 17,661 images**, which meets the threshold.

1.5 Proposal and approval status

The dataset proposal (A3-M1) was submitted with this protocol. At the time of writing no formal decision had been returned by the instructor. The group checked the proposal against the handbook rules (Section 28: genuine pairs, about 10,000 pairs for VQA, split by entity) and continued with the plan; any later condition from the instructor will be added to the assignment page.

1.6 Exploratory data analysis

Source: `Assignment3/01_eda.ipynb`.

1.7 Pair counts and missing modalities

Official split	Questions	Labelled	Distinct images	Images with 2 questions
train	17,056	17,056	16,539	517
validation	1,145	1,145	1,122	23
test	6,702	0	6,030	

Table 2: Official splits. Images are identified by the MD5 hash of their bytes, because the files have no image id.

Check on the 18,201 labelled questions	Result
Missing or undecodable image, empty question	0
Not exactly 4 choices, label outside 0–3	0
Not 10 direct answers or not 3 rationales	0
Questions with an empty choice	7 (0.04%)
Questions with two identical choices	37 (0.20%)
Images shared by the official train and validation splits	0
Grayscale images (converted to RGB)	20 of 17,661

Table 3: Missing-modality and validity checks.

No image or question is missing, so no missing-modality strategy is needed for training. The 44 questions with an empty or repeated choice are kept; they are counted in the error analysis. Section 3.3 tests what happens when the image is replaced.

1.8 Images

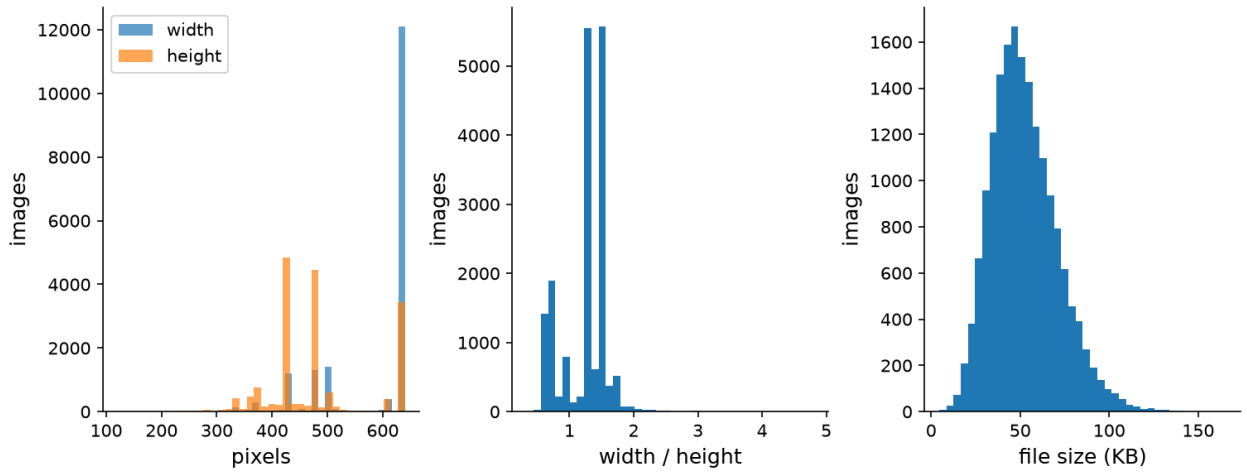


Figure 1: Original image width, height and file size.

The photos are COCO images, mostly 640×480 (4,005) or 640×427 (2,295); the median aspect ratio is 1.33 and the extremes are 0.33 and 4.81. Resizing to 224×224 squeezes a 4:3 photo horizontally by 25%. CLIP's own preprocessing resizes the short side and takes a centre crop instead; we kept the plain resize for all models and list the crop as future work.

1.9 Text

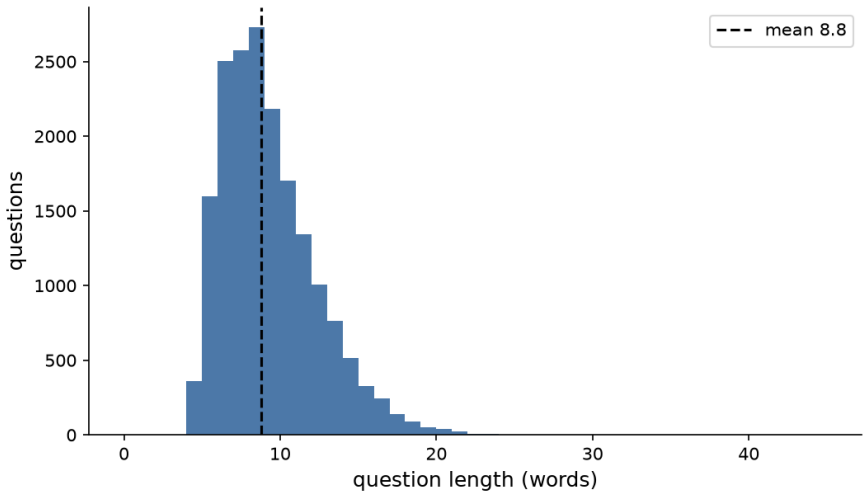


Figure 2: Question length in words.

Words	mean	median	max
question	8.8	8	44
choice	1.3	1	4
direct answer	1.2	1	4
rationale	11.1	9	65

Opening	Questions	Share
what is	4,109	22.6%
what type	1,102	6.1%
what are	850	4.7%
why is	730	4.0%
how many	595	3.3%
what color	542	3.0%
what kind	526	2.9%

Table 4: Text lengths (left) and the most common two-word question openings (right).

Questions are short (median 8 words) and every text fits the 77-token limit of the CLIP text encoder. The vocabulary is large (6,508 words in questions, 13,456 in choices) because the topics are open: animals, sports, food, history, brands. 64.5% of the questions start with “what” and 7.7% with “why”. A-OKVQA has no question-type field, so we use the opening words as a rough type in the error analysis.

1.10 Labels and answer agreement

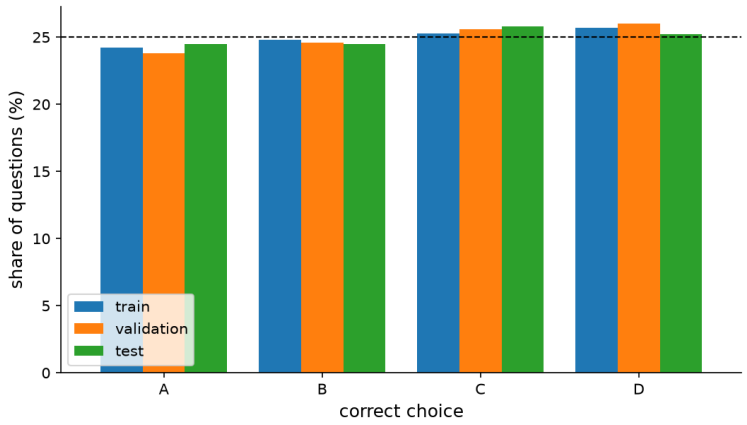


Figure 3: Share of each correct letter.

The correct letter is spread evenly: A 24.2%, B 24.8%, C 25.4%, D 25.7% over all labelled questions, so a letter bias cannot help. The correct choice appears among the 10 direct answers for every question, given on average by 3.7 of the 10 workers. Agreement between workers is moderate: the most common direct answer is given by 5.0 workers on average, and by 3 or fewer for 32.4% of the questions. The dataset flags 8.9% of the questions as difficult for direct answering.

1.11 Pair quality



Figure 4: Eight random training pairs (seed 42); * marks the correct choice and “Why” is the first rationale.

The examples show the mix the dataset was built for: counting (how many people on the scooter), reading the scene (the hour on the clock), and outside knowledge (which gang wears red, which civilisation made a vase). In 76.8% of the questions at least one rationale repeats a word of the correct choice, so the rationales usually talk about the answer, which makes them useful to explain errors later.

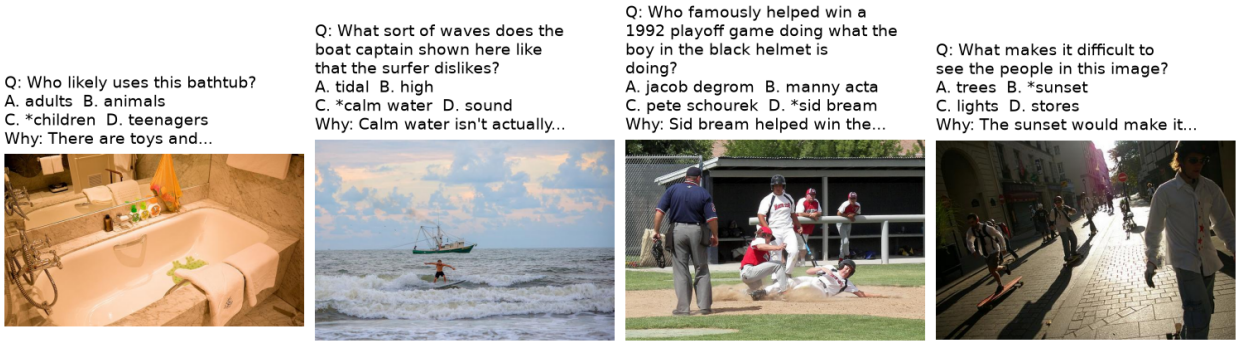


Figure 5: Four training questions with low agreement among the 10 direct answers (possible noise).

Two kinds of low agreement show up. Some questions are vague (who uses this bathtub: adults or children?), so several answers are reasonable. Others need rare knowledge (which player won a 1992 playoff game by sliding home), which

many workers did not know. In both cases the multiple-choice label is still one of the direct answers. We keep these questions; Section 3.7 compares accuracy on the high- and low-agreement subsets.

1.12 How much can text alone answer?

Simple rules that never look at the image, built on our train split and scored on our validation split (1,710 questions):

Rule	Validation accuracy
Chance	25.0%
Always the most frequent letter (D)	26.0%
Choice whose text was most often correct in training	31.1%
Choice sharing most words with the question	24.1%
Longest choice	28.5%

Table 5: Text-only rules.

The best rule reaches 31.1%, only 6 points above chance, although 89.6% of the validation questions offer at least one choice that was correct somewhere in training. The dataset was built so that the wrong choices are plausible answers, so answer frequency and word overlap help little. This is the opposite of datasets where the text alone gives most answers, and it means the image should matter. A trained text-only model learns more than these rules (43.3%, Section 3.2), but stays far below the models that see the image.

1.13 Splits and leakage prevention

Item	Plan
Test	The official validation split: 1,145 questions on 1,122 images, used once at the end
Train / validation	The official train split divided about 90/10 by image: 15,346 / 1,710 questions on 14,885 / 1,654 images
Seed	`GroupShuffleSplit`, seed 42, groups = image hash; saved in `Assignment3/splits.csv`
Split unit	The image. 540 images have two questions, and both always stay in the same split; the code asserts that no image is in two splits (0 overlaps found)
Unused	The official test split (6,702 questions, no labels)
Leakage notes	CLIP was pretrained on web images, and some COCO photos may be among them; this cannot be checked and affects all models equally
Subset rules	None. All 18,201 labelled questions are used

Table 6: Split plan.

The letter shares and the share of difficult questions are close in the three splits (difficult: 9.1%, 8.6% and 6.1%). With 1,145 test questions one standard error of accuracy is about 1.4 points, so small differences are judged with the five-seed comparison.

1.14 Preprocessing

- Image: decode JPEG, convert to RGB, resize to 224×224 (bicubic), scale to $[0, 1]$, normalise with the CLIP mean and standard deviation. No augmentation in the first runs.
- Text: the question and each choice go through the CLIP tokenizer (lower case, byte-pair encoding, at most 77 tokens). For the fusion models the text of choice j is "question + answer: choice$_j$".
- Batch: images $[B, 3, 224, 224]$ float32, labels $[B]$ int64, plus B questions and B lists of 4 choices. One batch of 32 has channel means $(-0.06, -0.09, -0.02)$ and standard deviations close to 1 after normalisation.

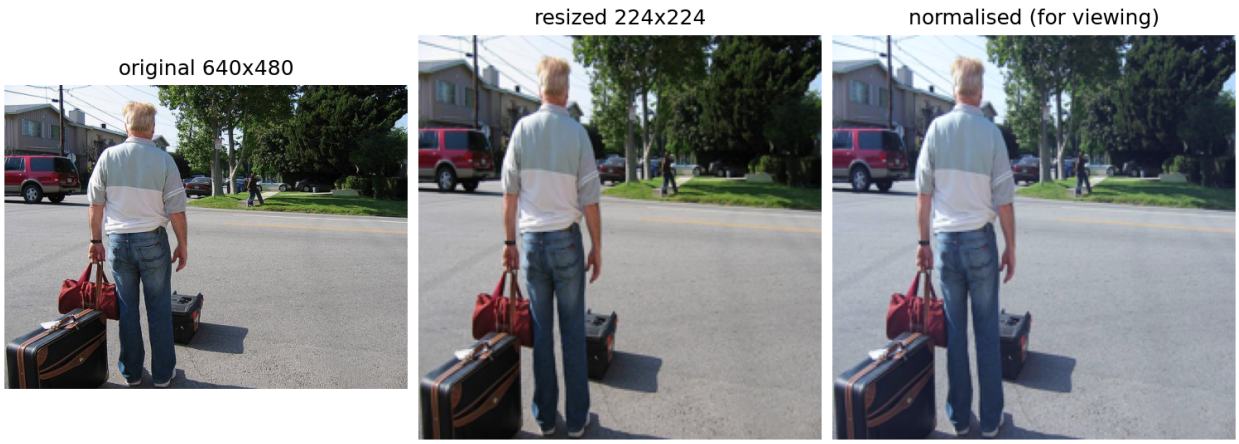


Figure 6: One training image before and after resizing and CLIP normalisation.

2 Methodology

2.1 Pipeline

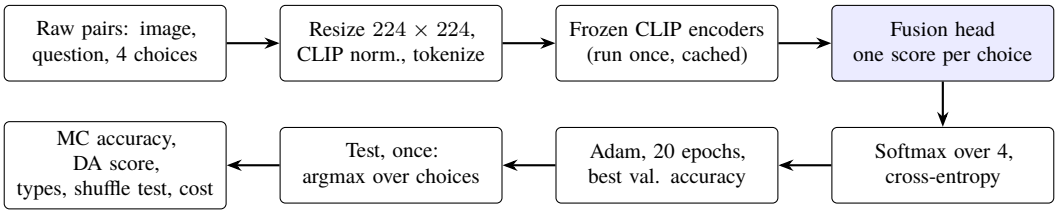


Figure 7: Pipeline: raw paired data, preprocessing, encoders, fusion, loss, optimisation, prediction and evaluation. Only the shaded fusion head changes between models.

Self-implemented vs library code. We wrote the data loader and split, the feature cache (`features.py`), all fusion heads (`models.py`), the training and evaluation loop with the shuffle test and the DA score (`train.py`), and the notebooks. Libraries: PyTorch (layers, `nn.MultiheadAttention`, Adam), Hugging Face `transformers` (CLIP model and tokenizer, pretrained weights), scikit-learn (`GroupShuffleSplit`), pandas, NumPy, Pillow and Matplotlib.

2.2 Encoders, representations and alignment

Both modalities use CLIP ViT-B/32 [3] (`openai/clip-vit-base-patch32`, the checkpoint of the course slides). CLIP was pretrained with a contrastive loss on 400M web image–text pairs, so its image and text vectors live in one 512-d space.

Modality	Encoder	Representation (cached, float16)
Image	ViT-B/32, 87.5M params, frozen	global vector $v \in \mathbb{R}^{512}$; 50 tokens $V \in \mathbb{R}^{50\times768}$ (class token + 7×7 patches); the input of the last block, for the fine-tune ablation
Question	text Transformer, 63.2M params, frozen	global vector $q \in \mathbb{R}^{512}$
Choice j	same text encoder	global vector $c_j \in \mathbb{R}^{512}$
"question answer: choice$_j$"	same text encoder	global vector $g_j \in \mathbb{R}^{512}$; tokens $T_j \in \mathbb{R}^{51\times512}$ with a padding mask

Table 7: Encoders and representations. 51 is the longest tokenized "question answer: choice" text.

Alignment. Every question belongs to exactly one image, so pairs are aligned by the dataset itself; `features.py` stores one row per distinct image (17,661) and each question points to its image index. Global vectors are scaled to length 1 before fusion, because CLIP compares vectors by direction. Since the encoders are frozen, their outputs are computed once and reused by every model and seed.

Missing-modality handling. No image or question is missing (Section 1), so no special handling is needed during training. Robustness to a wrong image is tested instead (Section 3.3).

2.3 Models and fusion

Each model gives one score s_j per choice; $p = \text{softmax}(s_1, \ldots, s_4)$. Every head is a small MLP: Linear → ReLU → Dropout(0.1) → Linear(512 → 1).

Model	Fusion stage	Input of the MLP for choice j	Trainable
Zero-shot CLIP	none	no MLP: $s_j = \cos(v, g_j)$	0
TextOnly (A)	none	$[q;\ c_j;\ q \odot c_j]$	787,457
ImageOnly (B)	none	$[v;\ c_j;\ v \odot c_j]$	787,457
LateFusion	late (global vectors)	$[v;\ q;\ c_j;\ v \odot c_j;\ q \odot c_j]$	1,311,745
CrossAttention	intermediate (tokens)	LateFusion input $+\ [r_j;\ r_j \odot g_j]$, with r_j from cross-attention	3,283,969

Table 8: Models. $\odot$ is the element-wise product, $[\cdot;\cdot]$ concatenation.

Fusion operation. Late fusion concatenates the global vectors and adds two element-wise products. The products let the MLP compare the image (or the question) with each choice dimension by dimension; Section 3.6 shows they matter most.

Improved fusion: cross-attention. The text tokens of "question answer: choice$_j$" are the queries, and the 50 image tokens are the keys and values, so each word can look at the image regions it needs. The attended tokens are mean-pooled into one region vector r_j per choice.

```
image = self.image_proj(self.image_norm(v_tokens))
image = image.repeat_interleave(4, 0)
queries = self.text_norm(qc_tokens.reshape(b * 4, L, 512))
attended, _ = self.attention(queries, image, image)
attended = self.norm(queries + attended)
region = masked_mean(attended, mask).view(b, 4, 512)
parts = [v, q, c, v * c, q * c,
         unit(region), unit(region) * qc]
scores = self.head(torch.cat(parts, -1)).squeeze(-1)
```

```
v_tokens              [B, 50, 768]
| LayerNorm, Linear   [B, 50, 512]
| repeat per choice   [B*4, 50, 512]
qc_tokens             [B, 4, 51, 512]
| LayerNorm           [B*4, 51, 512]
| attention, 8 heads  [B*4, 51, 512]
| residual + norm     [B*4, 51, 512]
| masked mean         [B, 4, 512]
| concat 7 parts      [B, 4, 3584]
| MLP 3584->512->1    [B, 4]
▼
scores                [B, 4]
```

CrossAttention keeps all late-fusion inputs and only adds the region branch, so a comparison with LateFusion isolates the effect of token-level fusion. This design was chosen on the validation split with seed 42: a first version that used only the region vector, without the late-fusion inputs and without LayerNorm on the tokens, reached 51.2% validation accuracy, against 58.9% for the final version (saved seed-42 run).

2.4 Loss, training and fine-tuning strategy

Loss. One loss for every model: cross-entropy over the 4 choices, $\ell = -\log p_y$. The unimodal baselines are trained on their own with the same loss; there is no extra per-modality loss in the fusion models.

Setting	Value
Optimizer	Adam, learning rate 10^{-3} for the heads, 10^{-5} for the CLIP block in the fine-tune ablation; no weight decay, no scheduler
Batch size, epochs	64, 20 epochs; every epoch runs (no early stopping)
Regularisation	Dropout 0.1 in every head and in the attention
Checkpoint rule	Highest validation accuracy; earliest epoch on a tie. Test used once, after the choice
Seeds	42–46 (PyTorch, NumPy, Python); the split is fixed (seed 42)
Pretraining / fine-tuning	CLIP pretrained and frozen; only the heads are trained. Ablation: the last CLIP vision block (7.1M parameters) is also trained, recomputed from the cached input of that block
Hardware	One NVIDIA A40 per process, float32 heads (features stored in float16)
Hyperparameter selection	Fixed in the proposal; only the cross-attention design was chosen on validation (seed 42)

Table 9: Training settings (`configs/a3.yaml`).

Because the last CLIP block is recomputed from its cached input, the fine-tune ablation gives exactly the same image features as CLIP at initialisation (checked: maximum difference 0).

2.5 Metrics

- **MC accuracy** (primary): the chosen option equals the correct option.
- **DA score**: $\min(n/3,\ 1)$, where n is the number of the 10 workers whose free-form answer equals the chosen text; partial credit for acceptable answers.
- **Shuffled-image accuracy**: test accuracy when every test image is replaced by the image of another test question (fixed permutation per seed).
- Breakdowns by question type (first words), difficult flag and worker agreement.
- Cost: parameters, training time, head time per question, feature caching time.
- Aggregation: mean $\pm$ sample SD over five seeds.

2.6 Experiment design

Experiment	Hypothesis	Changed factor	Fixed factors
E1 unimodal vs multimodal (M2)	Each modality alone is worse than both together	Modalities given to the head (text, image, both)	Features, head size, optimizer, epochs, seeds, split
E2 fusion level (M3)	Token-level cross-attention beats late fusion of global vectors	LateFusion vs CrossAttention	Same as E1
E3 fusion operation	The product terms are needed	LateFusion with and without $\odot$	Same as E1
E4 freeze vs fine-tune	Training the last CLIP block helps	Last vision block frozen or trained	Same as E1
E5 shuffled images	The fusion models use the image	Test images replaced by other images	Trained checkpoints

Table 10: Experiments. Metric: test MC accuracy (DA score as a check). Decision rule for every comparison, fixed in the proposal: a difference is supported if the gap between the two means is larger than the larger of the two seed SDs.

2.7 Reproduction and recorded environment

```
cd Assignment3
python download.py          # A-OKVQA parquet files, revision d1b0efa
python data_loader.py       # splits.csv (grouped by image, seed 42)
python features.py          # CLIP features of 17,661 images and 163,809 texts (one GPU, ~3 min)
python train.py             # 7 configurations x seeds 42-46 -> outputs/<run>_seed<s>_<time>/
jupyter nbconvert --to notebook --execute --inplace 02_results.ipynb
```

Each run folder keeps `best.pt`, `metrics.json` (settings, commit, environment), `history.csv` and the test scores. Environment: Python 3.12.16, PyTorch 2.14.0 (CUDA 13.0), `transformers` 5.22.2, NVIDIA A40.

3 Implementation results

Source: `Assignment3/02_results.ipynb`. 7 configurations × seeds 42–46 = 35 runs, each on one NVIDIA A40. Values are mean ± SD over the five seeds unless stated otherwise.

3.1 Training behaviour

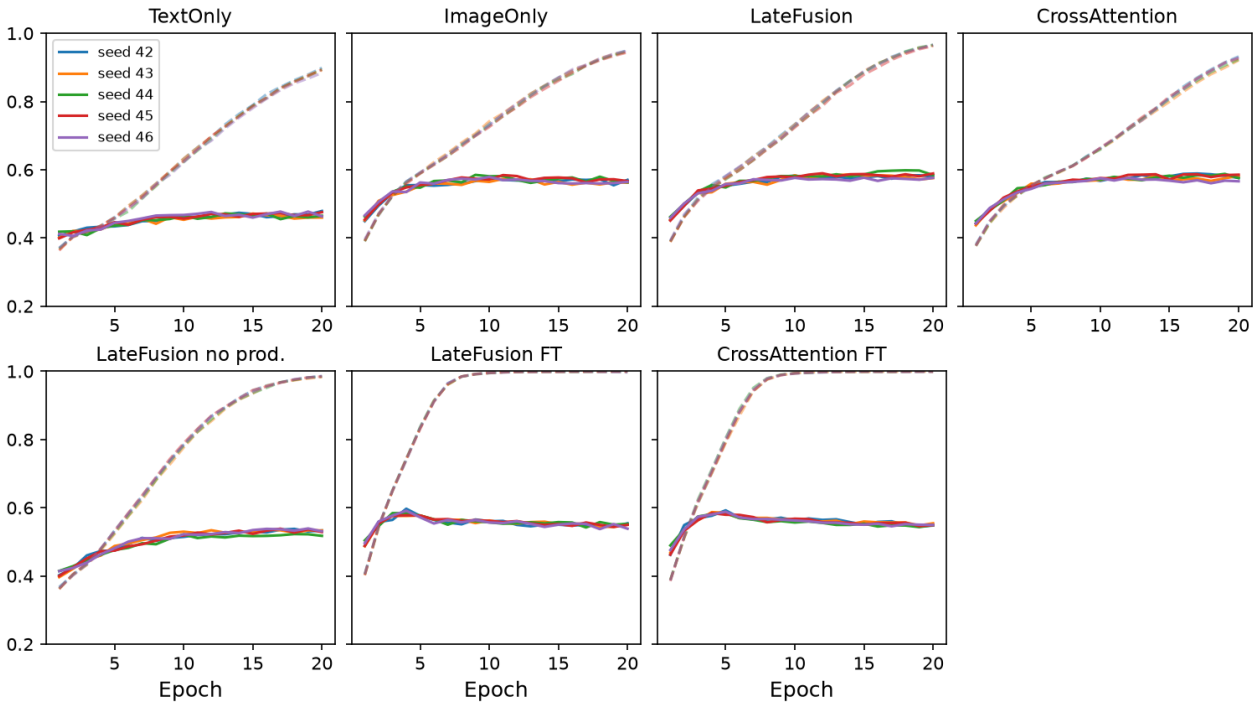


Figure 8: Validation (solid) and training (dashed) accuracy per epoch, one colour per seed.

Model	Selected epoch (seeds 42–46)	Comment
TextOnly	20, 15, 13, 20, 17	slow, still rising for two seeds
ImageOnly	14, 11, 9, 11, 10	
LateFusion	19, 14, 18, 12, 15	
CrossAttention	17, 14, 19, 14, 12	
LateFusionNoProduct	18, 16, 18, 17, 19	
LateFusionFineTune	4, 5, 3, 4, 4	overfits early
CrossAttentionFineTune	5, 4, 5, 4, 5	overfits early

Table 11: Checkpoint selected by validation accuracy.

All heads overfit: training accuracy of the fusion models passes 0.9, while validation accuracy stays near 0.58. With frozen features the best epoch is 9 to 20, so 20 epochs is enough budget. When the last CLIP block is also trained (7.1M more parameters) the best epoch is 3 to 5 and validation accuracy then falls, so the extra capacity is used to memorise the 15,346 training questions rather than to generalise. **Implementation issues.** (1) The first cross-attention design learned slowly (51.2% validation); normalising the tokens and keeping the late-fusion inputs fixed it (Section 2.3). (2) The CLIP encoder layer in `transformers` 5.22 takes two positional arguments; we checked the recomputed last block against the full model (difference 0). (3) The parquet files store `direct_answers` as a string, parsed back with `ast.literal_eval`.

3.2 Main comparison: unimodal vs multimodal

Model	Val. accuracy	Test accuracy	Test DA score	Trainable
Chance		0.250		0
Answer-text prior		0.312	0.281	0
Zero-shot CLIP		0.536	0.438	0
TextOnly (A)	0.475 ± 0.003	0.433 ± 0.010	0.362 ± 0.008	0.79M
ImageOnly (B)	0.579 ± 0.005	0.572 ± 0.010	0.463 ± 0.009	0.79M
LateFusion	**0.587** ± 0.008	**0.577** ± 0.009	**0.470** ± 0.008	1.31M
CrossAttention	0.584 ± 0.005	0.576 ± 0.007	0.470 ± 0.005	3.28M

Test accuracy	42	43	44	45	46
TextOnly	0.4201	0.4419	0.4271	0.4332	0.4445
ImageOnly	0.5546	0.5764	0.5773	0.5738	0.5790
LateFusion	0.5616	0.5843	0.5799	0.5825	0.5764
CrossAttention	0.5712	0.5677	0.5834	0.5817	0.5755

Table 12: Main models (top: mean ± SD over 5 seeds; bottom: per seed). The references need no training.

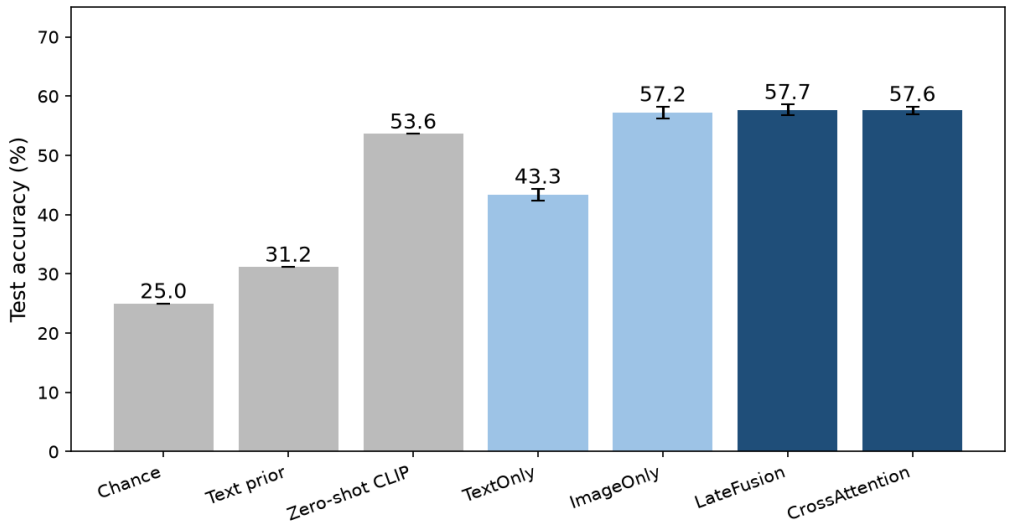


Figure 9: Test accuracy of the references and of the four main models (error bars: seed SD).

E1 decision. Adding the image to the text is a large and clear gain: LateFusion is 14.4 ± 0.8 points above TextOnly, in 5 of 5 seeds (supported). Adding the question to the image is not: LateFusion is only 0.5 ± 0.5 points above ImageOnly, below the seed SD, and ImageOnly wins in one seed (not supported). So the multimodal model beats modality A by far, but is only level with modality B. For CLIP features, the 4 choices plus the image carry most of the information. Our hypothesis is that the wrong choices are often things that are not in the picture, so matching each choice with the image already finds the answer; we did not check this by hand. The question then adds little, except for the types in Section 3.4. Zero-shot CLIP, which only matches the image with "question answer: choice" and is never trained, already reaches 53.6%; the trained heads add about 4 points.

3.3 Does the model use the image? Shuffled images

Model	Test accuracy	With shuffled images	Drop (points)
TextOnly	0.433 ± 0.010	0.433 ± 0.010	0.0
ImageOnly	0.572 ± 0.010	0.322 ± 0.005	25.0
LateFusion	0.577 ± 0.009	0.343 ± 0.003	23.4
CrossAttention	0.576 ± 0.007	0.345 ± 0.011	23.1
LateFusionNoProduct	0.506 ± 0.005	0.354 ± 0.002	15.1
LateFusionFineTune	0.575 ± 0.010	0.314 ± 0.011	26.1
CrossAttentionFineTune	0.582 ± 0.009	0.323 ± 0.013	26.0

Table 13: E5: every test image replaced by the image of another test question.

With a wrong image every image-based model falls to 31–35%, a drop of 23 to 26 points (15 for the model without products). TextOnly does not change. The fusion models therefore depend on the image, and a wrong image pulls them *below* the text-only model (43.3%): the image is not an optional hint but the main evidence.

3.4 Results by question type

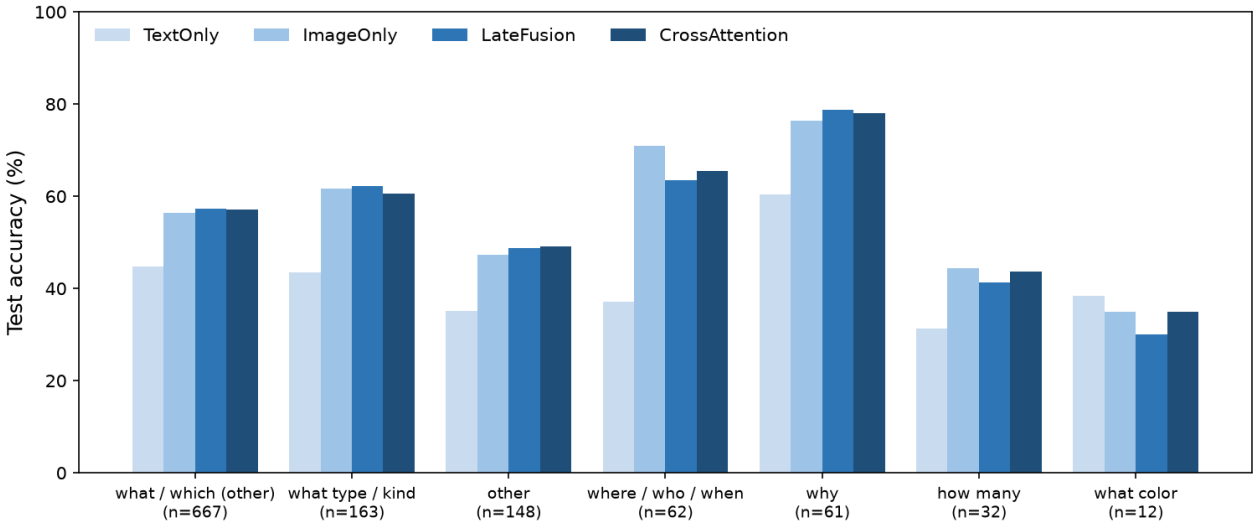


Figure 10: Test accuracy by question opening (mean over seeds); n is the number of test questions.

Type	Questions	TextOnly	ImageOnly	LateFusion	CrossAttn.	Zero-shot
what / which (other)	667	0.448	0.563	0.574	0.572	0.547
what type / kind	163	0.434	0.617	0.622	0.606	0.564
other	148	0.351	0.473	0.488	0.492	0.473
where / who / when	62	0.371	0.710	0.635	0.655	0.613
why	61	0.603	0.764	0.787	0.780	0.590
how many	32	0.312	0.444	0.412	0.438	0.188
what color	12	0.383	0.350	0.300	0.350	0.583
difficult (flag)	70	0.349	0.566	0.537	0.583	

Table 14: Test accuracy by question type, averaged over seeds.

The image helps most on where / who / when (+26.5 points over TextOnly), what type / kind (+18.8) and why (+18.4). Why questions are the easiest (79%): the choices are explanations, the text alone already gets 60%, and fusion is 2.3 points above image only, one of the few types where the question adds something. Counting stays weak (41–44%) because one global vector does not count objects, and zero-shot CLIP is even worse (19%). What color has only 12 questions; the trained heads (30–35%) fall below zero-shot CLIP (58%), which hints that the heads lose some colour information, but 12 questions are too few to conclude. On the 70 difficult questions CrossAttention is 4.6 points above LateFusion, the only place where token-level fusion looks better; with 70 questions this is a hint, not a result.

3.5 Support and conflict between modalities

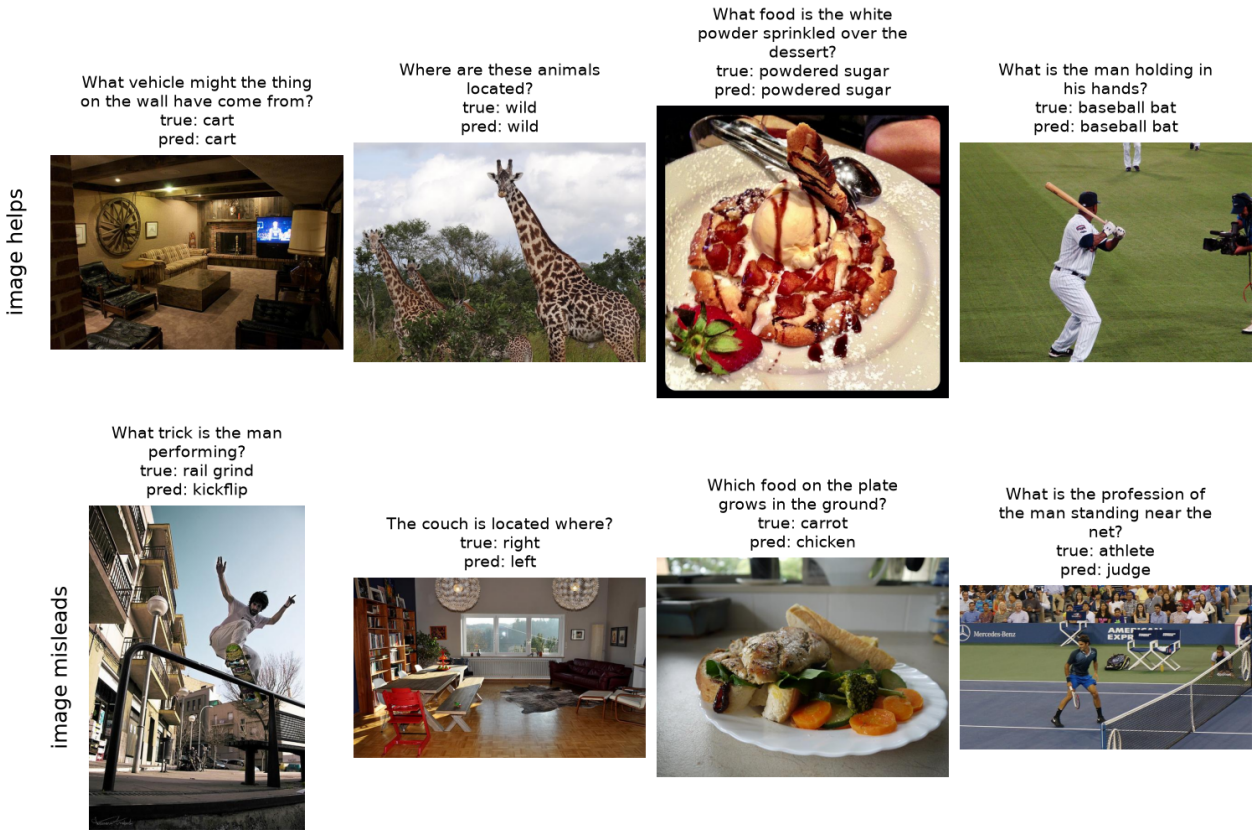


Figure 11: Seed 42. Top: the image helps (LateFusion right, TextOnly wrong). Bottom: the image misleads (TextOnly right, LateFusion wrong).

For seed 42 the image helps on 263 test questions and misleads on 101 (Table: 380 questions both right, 401 both wrong). Over the five seeds support is 254–282 and conflict 86–111, about 2.6 support cases for each conflict case. Support cases ask about something clearly visible (a baseball bat, powdered sugar, wild animals). Conflict cases share a pattern: the image contains a plausible but wrong answer. The model picks “left” for a couch on the right (a global vector has no position), “chicken” instead of “carrot” for the food that grows in the ground (both are on the plate), and “judge” for a tennis player near the net.

3.6 Ablation study

Ablation	Model	Test accuracy	Paired change	Seeds better	Decision
Reference	LateFusion	0.577 ± 0.009			
E2 fusion level	CrossAttention	0.576 ± 0.007	-0.1 ± 1.0	2/5	not supported
E3 fusion operation	LateFusionNoProduct	0.506 ± 0.005	-7.1 ± 0.7	0/5	products needed
E4 fine-tune (late)	LateFusionFineTune	0.575 ± 0.010	-0.2 ± 1.7	2/5	not supported
E4 fine-tune (cross)	CrossAttentionFineTune	0.582 ± 0.009	$+0.6 \pm 1.3$	3/5	not supported
E1 remove image	TextOnly	0.433 ± 0.010	-14.4 ± 0.8	0/5	image needed
E1 remove question	ImageOnly	0.572 ± 0.010	-0.5 ± 0.5	1/5	not supported

Table 15: Ablations, each against its base model (CrossAttentionFineTune against CrossAttention). Paired change in points, mean $\pm$ SD over seeds.

- **Fusion operation (E3)**: removing the element-wise products costs 7.1 points in every seed, the largest effect of any design choice. Concatenation alone makes the MLP learn the comparison between image and choice itself, which it does poorly with 15k examples.
- **Fusion level (E2)**: token-level cross-attention gives the same accuracy as late fusion, at 2.5 times the parameters and 9 times the training time per epoch.
- **Freeze vs fine-tune (E4)**: training the last CLIP block changes accuracy by less than one seed SD and makes the models overfit after 3–5 epochs.

3.7 Error analysis

The best fusion model by validation accuracy is CrossAttentionFineTune (0.587 validation, 0.582 test). Its errors, averaged over seeds:

Question type	Errors per seed	Questions	Error rate
what / which (other)	286.6	667	0.430
other	72.2	148	0.488
what type / kind	60.2	163	0.369
where / who / when	20.4	62	0.329
how many	16.6	32	0.519
why	13.2	61	0.216
what color	9.0	12	0.750

Table 16: Errors of CrossAttentionFineTune by question type. Accuracy is 54.1% on the 313 low-agreement questions and 59.8% on the 832 high-agreement ones.

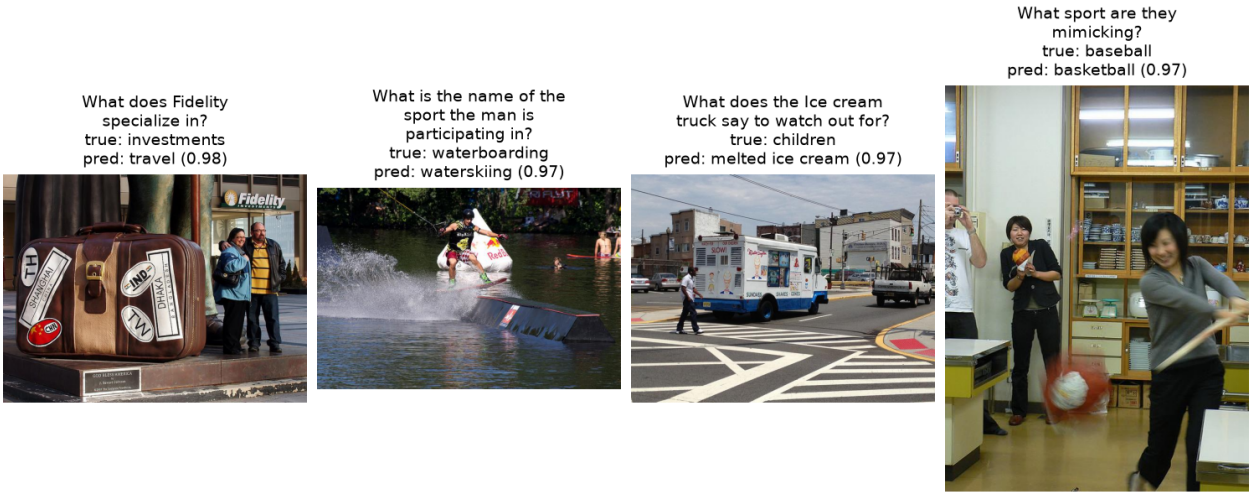


Figure 12: The four most confident errors of CrossAttentionFineTune, seed 42 (probability of the wrong choice in brackets).

Error group	Description and frequency	Hypothesised cause	Improvement idea
Visible distractor	The most visible object is chosen instead of the asked one (suitcase → "travel" for Fidelity; chicken vs carrot)	One global image vector; choices are matched to the whole image	Region features or the cross-attention branch with more data; use the rationale as extra supervision
Text in the image	Answers written on signs ("watch out for children" on a truck)	CLIP features do not read text reliably	Add an OCR model as a third input
Counting	how many: error rate 52%	Global features do not count	Object detector counts as input
Position	left / right questions	No position in global vectors; images are squeezed	Patch tokens with position; keep aspect ratio
Colour	what color: 75% errors (12 questions)	Head loses colour detail that zero-shot CLIP keeps	Mix zero-shot CLIP scores into the head
Ambiguous label	313 low-agreement questions: 54.1% vs 59.8%	Workers disagree; several answers acceptable	Report the DA score next to MC accuracy

Table 17: Error groups. Frequencies are from the tables above; causes are hypotheses, not tested here.

3.8 Compute cost

Model	Trainable	s / epoch	Train min	Head ms / question	Test acc.
TextOnly	0.79M	0.29	0.10	0.002	0.433
ImageOnly	0.79M	0.29	0.10	0.002	0.572
LateFusion	1.31M	0.32	0.11	0.003	0.577
LateFusionNoProduct	0.79M	0.31	0.11	0.002	0.506
CrossAttention	3.28M	2.74	0.92	0.056	0.576
LateFusionFineTune	8.79M	2.85	0.95	0.053	0.575
CrossAttentionFineTune	10.77M	5.52	1.84	0.107	0.582

Table 18: Cost on one NVIDIA A40 with cached features. One-time feature caching: 99 s for 17,661 images (5.6 ms per image) and 72 s for 163,809 texts. The frozen CLIP encoders have 151M parameters in total.

The cost is dominated by the frozen CLIP encoders, which every model shares; once the features are cached, a head trains in 6 seconds (LateFusion) to 2 minutes and answers a question in well under 0.1 ms. LateFusion gives the best accuracy per unit of cost; CrossAttention costs 9 times more per epoch for the same accuracy.

3.9 Comparison and discussion

Which model is better, and by how much. All fusion models land at 57.5–58.2% test accuracy, within one seed SD of each other. They are 14 points above text only, 4 points above zero-shot CLIP and 33 points above chance, but only about 0.5 points above the image-only model. The planned improvement (token-level cross-attention) did not beat simple late fusion, and fine-tuning the last CLIP block did not either. The handbook allows this when it is explained; the evidence points to three causes. (1) *Data*: image–choice matching already solves most questions (ImageOnly 57.2%, zero-shot CLIP 53.6%), so there is little left for the question or for region-level fusion to add. (2) *Size*: 15k training questions are few for learning attention over image regions; the fine-tuned models overfit within 5 epochs. (3) *Features*: CLIP was trained to match the pooled class token with text, not the 50 patch tokens, so these tokens may not be well suited to region-level matching.

Modality interaction. The image is the main evidence: removing it costs 14.4 points, and a wrong image costs 23–26 points, pulling the model below text-only accuracy. Support cases outnumber conflict cases 2.6 to 1. Conflicts happen when the image contains a plausible wrong answer, which a model that matches choices with the whole image cannot reject.

What cannot be concluded. Per-type differences for small groups (12 colour, 32 counting, 70 difficult questions) are within noise. We did not test other encoders, so we cannot say whether a stronger encoder or region features would let token-level fusion help.

3.10 Limitations and conclusion

Limitations. One frozen encoder (CLIP ViT-B/32) and short training; no search beyond the fixed proposal settings, except the cross-attention design chosen on validation. Images are squeezed to 224×224. The test split is the official validation split (1,145 questions), because the official test labels are not public. Question types come from the first words, not from an official label. The proposal had no formal approval decision when the experiments ran (Section 1).

Future work. Region features or an object detector for counting and position, OCR for text in images, the rationales as extra training signal, and a stronger encoder (for example a larger CLIP or a VLM) with the same protocol.

Answer to the research question. The image is the modality that matters: it adds 14.4 points over text alone, and replacing it with a wrong image costs 23–26 points. The question adds only 0.5 points once the image and the choices are known. The way the vectors are combined matters more than where: element-wise products add 7.1 points, while token-level cross-attention and fine-tuning the last CLIP block do not improve on late fusion (about 58% test accuracy for all).

4 AI Usage Disclosure

The group used generative AI tools as disclosed below and in `AI_USAGE.md`. All numbers in this report are computed by the committed notebooks from saved run outputs.

Field	Entry
Tool	Claude Code (Anthropic), model Claude Opus 5.5, with helper agents on Claude Sonnet, in the group's VS Code workspace
Used by	Võ Thị Như Quỳnh
Time / stage	9 November 2026: dataset search, proposal, M2 and M3
Purpose	Coding assistance (`download.py`, `data_loader.py`, `features.py`, `models.py`, `train.py`, notebooks); running the 35 training runs; first drafts of the M2 and M3 reports, the slides and the assignment pages. The dataset and the decision to continue before a formal approval were made by the group
Affected sections	All sections of this report; `Assignment3/`; `configs/a3.yaml`
Representative prompt	"Continue with Assignment 3 M2 and M3 now; follow the handbook and the style of Assignment 1; keep the code simple."
How output was verified	Every number is read from `metrics.json` and prediction files by `02_results.ipynb`; the recomputed CLIP block was checked against the full model; the group reviews code and text before submission
Responsible member	Võ Thị Như Quỳnh
Verification sources	A-OKVQA and CLIP papers, course slides (multimodal learning: CLIP, cross-attention), PyTorch and `transformers` documentation

Table 19: AI usage for Assignment 3.

All members are able to explain every part of the submission and take responsibility for the submitted code, numbers, figures and text.

References

[1] D. Schwenk, A. Khandelwal, C. Clark, K. Marino and R. Mottaghi, "A-OKVQA: A benchmark for visual question answering using world knowledge," in *Proc. ECCV*, pp. 146–162, 2022. Dataset: https://huggingface.co/datasets/HuggingFaceM4/A-OKVQA, revision d1b0efa.

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[3] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. ICML*, PMLR 139, pp. 8748–8763, 2021.

[4] A. Vaswani et al., "Attention is all you need," in *Advances in Neural Information Processing Systems 30*, pp. 5998–6008, 2017.

[5] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. ICLR*, 2015.